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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Human Research Program Risks:	None		
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Contact Monitor:		**Contact Phone:**	
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	Skeletal unloading results in an inhibition of bone formation associated with a decrease in osteoblast number, impaired mineralization of bone, and altered proliferation and differentiation of osteoprogenitor cells. Although such changes are likely to be mediated by multiple factors, resistance to the skeletal growth promoting action of insulin like growth factor-I (IGF-I) appears to play an important role. IGF-I mRNA and protein levels rise during skeletal unloading, but the ability of IGF-I to stimulate proliferation of bone cells in vivo and in vitro when taken from unloaded bone is markedly impaired. This resistance to IGF-I is associated with a failure of IGF-I to activate its receptor and the subsequent IGF-I signaling pathways. BMSC from unloaded bone show a reduction in integrin expression, and inhibition of integrin signaling reproduces the findings of IGF-I resistance. Therefore, we propose to test the following hypothesis: Skeletal unloading inhibits the ability of IGF-I to stimulate osteoprogenitor cell proliferation by impairing, via an		

Task Description:	integrin-mediated mechanism, IGF-I signal transduction. This hypothesis will be tested by achieving three specific aims: 1) Determine the points at which skeletal unloading blocks IGF-I stimulated bone formation. This will be accomplished by determining whether unloading alters stem cell commitment to the osteoblast lineage, IGF-I receptor activation, and/or the mRNA and protein levels and/or activation of the components of the IGF-I signaling pathway(s). 2) Determine the ability of skeletal unloading to block the response of osteoprogenitor cells to other growth factors such as EGF, PDGF, and FGF with respect to proliferation and activation of the MAPK and/or PI3K pathway. Those growth factors found to affected similarly to IGF-I by skeletal unloading will be further explored with respect to mechanism. 3) Determine the interaction between IGF-I and integrin signaling in the regulation of osteoprogenitor cell proliferation. This will be accomplished by determining the effect of skeletal unloading on the mRNA and protein levels of the various integrins in osteoprogenitor cells, determining which integrins potentiate IGF-I responsiveness, whether the effect of integrins on IGF-I signaling is direct or indirect, and whether correction of integrin levels, if altered by skeletal unloading, will restore IGF-I responsiveness. We anticipate that these studies will increase our understanding of the mechanism(s) for bone loss in weightlessness, and lead to measures that will prevent or reverse such losses. Specific Aims: 1. Determine the point(s) at which skeletal unloading blocks IGF-I stimulated bone formation. a. Determine whether skeletal unloading alters the mesenchymal pool committed to osteoblast differentiation. b. Determine whether skeletal unloading alters the ability of IGF-I to bind to and activate the IGF-I receptor (IGF-IR) in those cells committed to osteoblast differentiation c. Determine the relative contributions of the ras/MAPK and PI3K/Akt pathways to the proliferative response to IGF-I. d. Determine whether skeletal unloading alters the protein levels and/or activation of the components of the pathway(s) mediating the proliferative response to IGF-I. e. Determine whether skeletal unloading alters the nuclear translocation of ERK, PI3K, and Akt and cellular levels of cyclin D1, p21cip, and p27kip 2. Determine the ability of skeletal unloading to block the response of BMSC to other growth factors such as EGF, PDGF and FGF with respect to: a. Proliferation b. Growth factor receptor number, binding, and activation c. Activation of the ras/MAPK and/or PI3K/Akt pathways 3. Determine the interaction between IGF-I and integrin signaling as affected by skeletal unloading in the regulation of osteoprogenitor cell proliferation. a. Determine the mRNA and protein levels of the various integrins in osteoprogenitor cells from normally loaded and unloaded bone b. Determine which integrins potentiate IGF-I responsiveness with respect to proliferation and activation of the IGF-IR and the ras/MAPK and PI3K/Akt pathways c. Determine whether the effect of integrins on IGF-IR activation requires direct interaction with the IGF-IR d. Determine whether correction of integrin levels, if altered by skeletal unloading, will restore IGF-I responsiveness with respect to proliferation and activation of IGF-I signaling pathways. The bone loss associated with skeletal unloading is of great importance to individuals subjected to prolonged periods of skeletal unloading whether this be in microgravity, in a cast, or following neural injury or stroke. Determining the mechanism by which such bone loss occurs is the first step toward effective prevention and treatment. Our project is directed at understanding this mechanism.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal seeks to understand the mechanism(s) for bone loss during skeletal unloading. This is a major problem clinically for immobilized and aging patients. Bone is a modeling and remodeling tissue meaning that it is undergoing continual development. We expect such research to enable us to understand whether, and how, the development of cells and systems (in this case bone) depend on mechanical forces, the role of cellular structures (bone cells within their own matrix) in response to mechanical forces, how the bone and bone cells sense and use such forces, and how physical and chemical signals interact to determine biological structure and function (bone shape and strength). Such research will contribute to our knowledge of how osteoporosis develops whether it be by aging, by immobilization, or by drugs. As our understanding of the processes by which bone maintains its integrity is increased, we anticipate that our ability to develop preventive measures and treatment for osteoporosis will improve.
Task Progress:	We have made good progress toward achieving these aims of the task. 1. Determine the points at which skeletal unloading blocks IGF-I stimulation bone formation. In this aim we have demonstrated that skeletal unloading does not alter the binding of IGF-I to its receptor. We have used an analog of IGF-I, namely des IGF-I, which does not bind to IGF-binding proteins and demonstrated that like IGF-I itself binding to the bone marrow stromal cells (BMSC) from loaded and unloaded bones is equivalent. We have shown that activation of the IGF-I receptor via phosphorylation is the critical step that is impaired by skeletal unloading. The down stream events such as activation of the MAPK pathway and the subsequent proliferative response to IGF-I are impaired in BMSC from unloaded bones, but this follows the failure to activate the IGF-I receptor. We have shown that skeletal unloading reduces the expression of a number of integrins, and the use of the integrin inhibitor echistatin reproduces the effects of skeletal unloading on IGF-I responsiveness. Thus we believe that the loss of bone during skeletal unloading is due to IGF-I resistance caused by the failure of integrins to enable IGF-I to activate its receptor. In contrast when the animals are reloaded, we are able to restore IGF-I responsiveness, integrin expression, and accelerated bone formation. 2. Determine the ability of skeletal unloading to block the response of BMSC to other growth factors. We have examined in depth the effect of skeletal unloading on the resistance to PDGF. To our surprise, BMSC from unloaded bones respond quite well to PDGF with respect to activation of the PDGF receptor (via phosphorylation) and proliferation. Furthermore, the integrin antagonist echistatin, which blocks the response of BMSC to IGF-I, fails to block the response of BMSC to PDGF. Thus, the resistance to IGF-I caused by skeletal unloading appears to be specific to IGF-I and is not generalized to all other growth factors. 3. Determine the interaction between IGF-I and integrin signaling as affected by skeletal unloading. As noted above we have found that the expression of a number of integrins is reduced in the BMSC from unloaded bones. We are having some difficulty showing that the protein levels are reduced because of the lack of good antibodies to rat integrins. However we have developed methods to selectively knockout specific integrins to determine which if any are critical for BMSC response to IGF-I. These experiments are being performed in both rat BMSC and a human osteoblast cell line for which better integrin antibodies are available. Using this latter cell line, we have had success in showing that at least one of the integrins of interest binds directly to the IGF-I receptor, as shown by co-immunoprecipitation. Furthermore, we have developed an in vitro model using pulsatile fluid flow to test whether integrins per se can activate the IGF-IR when load is applied in the absence of IGF-I. Our results show that load can activate the IGF-IR, which appears to be dependent on selected integrins and the matrix containing selected integrin ligands on which the cells are plated.

Bibliography Type:	Description: (Last Updated: 04/15/2008)
Abstracts for Journals and Proceedings	Long RK, Bikle DD. "Fluid flow activation of ERK in bone marrow stromal cells requires the IGF-I receptor." Presented at the 28th annual meeting of the American Society for Bone and Mineral Research, Philadelphia, PA, Sept 15-19, 2006. J Bone Min Res 2006;21(suppl 1): S372. , Sep-2006
Abstracts for Journals and Proceedings	Wang Y, ElAlieh HZ, Chang W, Halloran BP, Clemens TL, Bikle DD. "Ablation of IGF-I signaling in osteoclast precursors impairs osteoclastogenesis." Presented at the 28th annual meeting of the American Society for Bone and Mineral Research, Philadelphia, PA, September 15-19, 2006. J Bone and Min Res 2006;21(suppl 1): S12. , Sep-2006
Abstracts for Journals and Proceedings	Wang Y, Clemens TL, Bikle DD, Chang W. "Chondrocyte specific knockout of IGF-IR impedes chondrocyte growth, survival, and differentiation." Presented at the 28th annual meeting of the American Society for Bone and Mineral Research, Philadelphia, PA, Sept 15-19, 2006. J Bone Min Res 2006;21(suppl 1): S50. , Sep-2006
Articles in Peer-reviewed Journals	Wang Y, Nishida S, Elalieh HZ, Long RK, Halloran BP, Bikle DD. "The role of IGF-I signaling in regulating osteoclastogenesis." J Bone Miner Res. 2006 Sep;21(9):1350-8. PMID: 16939393 , Sep-2006
Articles in Peer-reviewed Journals	Burghardt A, Wang Y, Elalieh H, Thibault X, Bikle D, Peyrin F, Majumdar S. "Evaluation of fetal bone structure and mineralization in IGF-I deficient mice using synchrotron radiation microtomography and Fourier transform infrared spectroscopy." 2007 Jan;40(1):160-8. Epub 2006 Aug 14. PMID: 16905376 , Jan-2007
Articles in Peer-reviewed Journals	Wang Y, Nishida S, Sakata T, Elalieh HZ, Chang W, Halloran BP, Doty SB, Bikle DD. "Insulin like growth factor-I is essential for embronic bone development." Endocrinology. 2006 Oct;147(10):4753-61. Epub 2006 Jul 20. PMID: 16857753 , Oct-2006
Articles in Peer-reviewed Journals	Bikle DD. "Integrins, insulin like growth factors, and the skeletal response to load." Osteoporos Int. 2008 Mar 29 [Epub ahead of print] PMID: 18373051 , Mar-2008